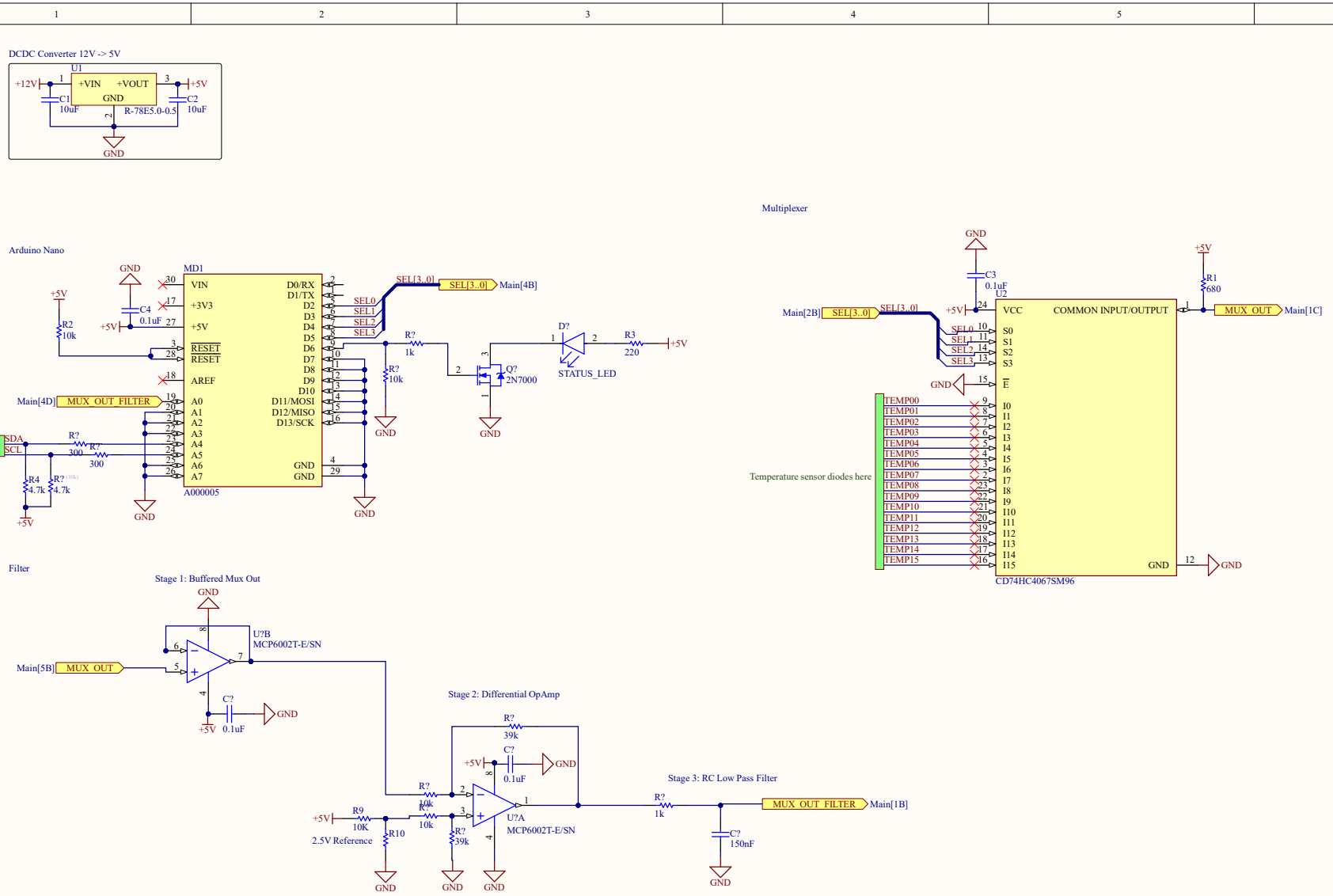

DCDC Converter 12V -> 5V
U1
+12V
+VIN
+VOUT
+5V
GND
C1
10uF
C2
10uF
R-78E5.0-0.5
Arduino Nano
MD1
VIN
+3V3
+5V
RESET
AREF
A0
A1
A2
A3
A4
A5
A6
A7
D0/RX
D1/TX
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11/MOSI
D12/MISO
D13/SCK
A000005
C4
0.1uF
R2
10k
Main[4D]
MUX_OUT_FILTER
I2C Output
SDA
SCL
R4
4.7k
SEL[3..0]
Main[4B]
SEL0
SEL1
SEL2
SEL3
STATUS_LED
2N7000
220
Multiplexer
Main[2B]
Main[1C]
MUX_OUT
COMMON INPUT/OUTPUT
VCC
S0
S1
S2
S3
E
I0
I1
I2
I3
I4
I5
I6
I7
I8
I9
I10
I11
I12
I13
I14
I15
TEMP00
TEMP01
TEMP02
TEMP03
TEMP04
TEMP05
TEMP06
TEMP07
TEMP08
TEMP09
TEMP10
TEMP11
TEMP12
TEMP13
TEMP14
TEMP15
CD74HC4067SM96
Temperature sensor diodes here
R1
680
C3
0.1uF
Filter
Stage 1: Buffered Mux Out
Main[5B]
U?B
MCP6002T-E/SN
Stage 2: Differential OpAmp
U?A
2.5V Reference
R9
10K
R10
Stage 3: RC Low Pass Filter
C?
150nF
Title
Size
B
Number
Revision
Date:
10/31/2026
Sheet of
File:
C:\Users\..\Main.SchDoc
Drawn By: